Post-Quantum Cryptography Suite for Zoo Network: ML-KEM, ML-DSA, SLH-DSA, and Ringtail

NIST FIPS 203/204/205 Integration

with Lattice-Based Threshold Signatures

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Abstract

We present the post-quantum cryptography suite deployed across Zoo Network, implementing all three NIST post-quantum standards—ML-KEM (FIPS 203), ML-DSA (FIPS 204), and SLH-DSA (FIPS 205)—alongside the Ringtail lattice-based threshold signature scheme. Each primitive serves a distinct role: K-Chain uses ML-KEM for key encapsulation and ML-DSA for transaction signatures; Q-Chain uses Ringtail for threshold validator operations; and the consensus layer uses hybrid BLS12-381 + Ringtail for dual-path finality. SLH-DSA provides hash-based stateless signatures for long-term key certification. GPU-accelerated Number Theoretic Transform (NTT) achieves 8x throughput over CPU. Hybrid classical+post-quantum composition ensures backward compatibility. All implementations are formally verified in Lean 4 (Crypto/MLKEM.lean, Crypto/MLDSA.lean, Crypto/SLHDSA.lean, Crypto/Ringtail.lean, Crypto/Hybrid.lean).

Contents

1 Introduction

1.1 The Quantum Threat

Blockchain cryptography faces two quantum-vulnerable primitive classes: key agreement (ECDH, broken by Shor's algorithm) and digital signatures (ECDSA, EdDSA, BLS, similarly broken). Hash functions and symmetric ciphers retain security with doubled key lengths. The "harvest now, decrypt later" model makes this threat immediate: transactions recorded today can be forged retroactively once quantum computers arrive [1].

1.2 NIST Standards and Zoo's Strategy

NIST finalized three post-quantum standards in 2024 [2]: ML-KEM (FIPS 203, key encapsulation, Module-LWE), ML-DSA (FIPS 204, digital signatures, Module-LWE), and SLH-DSA (FIPS 205, digital signatures, hash-based). Zoo deploys each where its properties are most advantageous, supplemented by Ringtail for threshold operations not covered by NIST.

1.3 Contributions

We contribute: (1) complete FIPS 203/204/205 integration into a production blockchain; (2) Ringtail threshold signatures with Ring-LWE security; (3) hybrid classical+PQ composition framework; (4) GPU NTT acceleration achieving 8x throughput; (5) Lean 4 formal verification of all primitives; (6) chain-specific deployment mapping across Zoo's multi-chain topology.

2 ML-KEM: Key Encapsulation (FIPS 203)

2.1 Construction

ML-KEM is based on the Module Learning With Errors (Module-LWE) problem. Zoo deploys ML-KEM-768 (NIST Security Level 3).

Definition 1 (Module-LWE)**.** Given $\mathbf{A} \xleftarrow{\$} R_q^{k\times k}$, $\mathbf{s} \leftarrow \beta_\eta^k$, $\mathbf{e} \leftarrow \beta_\eta^k$, distinguish $(\mathbf{A}, \mathbf{A}\mathbf{s} + \mathbf{e})$ from $(\mathbf{A}, \mathbf{u})$ where $\mathbf{u} \xleftarrow{\$} R_q^k$, with $R_q = \mathbb{Z}_q[X]/(X^{256} + 1)$, $q = 3329$, $k = 3$.

Key generation produces $pk = (\rho, \hat{\mathbf{t}})$ where $\mathbf{t} = \mathbf{A}\mathbf{s} + \mathbf{e}$. Encapsulation computes ciphertext via $\mathbf{u} = \mathrm{NTT}^{-1}(\hat{\mathbf{A}}^T \hat{\mathbf{r}}) + \mathbf{e}_1$ and $v = \mathrm{NTT}^{-1}(\hat{\mathbf{t}}^T \hat{\mathbf{r}}) + e_2 + \lceil q/2 \rfloor \cdot m$. Decapsulation with FO transform achieves IND-CCA2 security.

2.2 Performance and Usage

Parameter	**ML-KEM-768**	**ECDH (X25519)**
Public key	1,184 B	32 B
Ciphertext	1,088 B	32 B
KeyGen / Encaps / Decaps	32 / 40 / 42 μs	48 / 150 / 150 μs

Table 1: ML-KEM-768 vs classical ECDH

ML-KEM is faster than X25519 despite 37x larger keys. Zoo uses it for validator P2P encrypted channels, optional PQ-encrypted transaction memos, and secure share distribution during Ringtail DKG.

3 ML-DSA: Digital Signatures (FIPS 204)

3.1 Construction

ML-DSA (formerly Dilithium) uses Module-LWE and Module-SIS with "Fiat-Shamir with Aborts" rejection sampling. Zoo deploys ML-DSA-65 (NIST Security Level 3, $k = 6$, $\ell = 5$, $q = 8380417$).

Key generation: $\mathbf{A} \xleftarrow{\$} R_q^{k\times\ell}$, $\mathbf{s}_1 \leftarrow S_\eta^\ell$, $\mathbf{s}_2 \leftarrow S_\eta^k$, $\mathbf{t} = \mathbf{A}\mathbf{s}_1 + \mathbf{s}_2$. Signing samples $\mathbf{y} \leftarrow S_{\gamma_1}^\ell$, computes challenge c from $\mathbf{w}_1 = \text{HighBits}(\mathbf{Ay})$, and outputs $\mathbf{z} = \mathbf{y} + c\mathbf{s}_1$ if norm bounds are satisfied (otherwise resamples).

3.2 Performance and Usage

Parameter	**ML-DSA-65**	**ECDSA (secp256k1)**
Public key / Signature	1,952 B / 3,309 B	33 B / 64 B
Sign / Verify	680 / 250 μs	45 / 120 μs

Table 2: ML-DSA-65 vs classical ECDSA

ML-DSA-65 is the default transaction signature on K-Chain: user transfers, contract calls, staking, governance proposals. The 59x key and 52x signature expansion is mitigated by pruning after finality.

4 SLH-DSA: Hash-Based Signatures (FIPS 205)

4.1 Construction

SLH-DSA (formerly SPHINCS+) is a stateless hash-based scheme whose security reduces solely to hash function second-preimage resistance—no lattice or number-theoretic assumptions. Zoo deploys SLH-DSA-SHA2-192s.

The scheme uses a hypertree of $d = 7$ XMSS trees of height h/d ($h = 63$ total), with FORS few-time signatures at the leaves. This provides 2^{63} signing capacity per key pair with the most conservative security margin of any post-quantum scheme.

4.2 Performance and Usage

Parameter	**SLH-DSA-SHA2-192s**	**ML-DSA-65**
Public key / Secret key	48 B / 96 B	1,952 B / 4,032 B
Signature	16,224 B	3,309 B
Sign / Verify	280 ms / 6.2 ms	0.68 ms / 0.25 ms

Table 3: SLH-DSA: compact keys, large signatures, slow operations

Reserved for high-assurance, low-frequency operations: root CA key certification, DAO constitutional amendments, long-term validator identity binding, cross-chain bridge anchors.

5 Ringtail: Post-Quantum Threshold Signatures

5.1 Construction

NIST standards lack threshold signatures. Ringtail fills this gap with a (t, n)-threshold scheme based on Ring-LWE [3, 8], using parameters $d = 1024$, $q = 2^{32} - 5$ (NTT-friendly), $\sigma = 3.19$ (Gaussian width), $\beta = 2^{20}$ (norm bound).

5.2 Threshold Protocol

DKG: each party P_i samples polynomial $f_i(x) \in R_q[x]$ of degree $t - 1$, broadcasts commitments, and sends encrypted shares via ML-KEM. Collective public key: $PK = \sum_{i=1}^{n} C_{i,0}$.

Signing with t signers from $\mathcal{T}$: each P_i computes partial signature $\sigma_i = \lambda_i \cdot sk_i \cdot H_{\text{lat}}(M) + e_i$ with Lagrange coefficients λ_i and fresh noise e_i. Aggregation: $\sigma = \sum_{i \in \mathcal{T}} \sigma_i$. Verification checks $\|\sigma\|_\infty < t \cdot \beta$ and the algebraic relation against PK.

5.3 Sizes

Component	Size (bytes)
Public key / Secret key share	4,096 / 4,096
Signature share / Threshold signature	4,096 / 4,096
DKG commitment / DKG share (encrypted)	4,096 / 5,280

Table 4: Ringtail object sizes ($d = 1024$, $q = 2^{32} - 5$)

6 Hybrid Classical + Post-Quantum Composition

6.1 Signature Composition

Hybrid signatures concatenate classical and PQ components:

$$\Sigma_{\text{hybrid}} = (\sigma_{\text{classical}}, \sigma_{\text{PQ}}, \text{tag}) \tag{1}$$

Verification requires **both** valid: $\text{Verify}_{\text{hybrid}} = \text{Verify}_{\text{classical}} \wedge \text{Verify}_{\text{PQ}}$. An adversary must break both to forge.

6.2 Key Encapsulation Composition

For key exchange, Zoo combines X25519 and ML-KEM-768:

$$K_{\text{shared}} = \text{HKDF-SHA256}(K_{\text{X25519}} \| K_{\text{ML-KEM}}, \text{info}, 32) \tag{2}$$

Secure if either X25519 or ML-KEM remains unbroken.

6.3 Transition Timeline

Phase 1 (current): hybrid mode, both required. Phase 2 (2028): PQ-primary, classical optional. Phase 3 (2030+): PQ-only via DAO governance vote.

7 GPU NTT Acceleration

All lattice primitives use polynomial multiplication in $R_q = \mathbb{Z}_q[X]/(X^n + 1)$ via NTT:

$$\mathbf{a} \cdot \mathbf{b} \bmod (X^n + 1) = \mathrm{NTT}^{-1}(\mathrm{NTT}(\mathbf{a}) \circ \mathrm{NTT}(\mathbf{b})) \tag{3}$$

The NTT's $\log_2(n)$ butterfly stages are fully parallelizable. For $n = 1024$: 10 stages, $n/2 = 512$ independent butterflies per stage, fitting in one GPU block (512 threads). Twiddle factors stored in constant memory.

Primitive	n	**CPU** (μ**s**)	**GPU** (μ**s**)	**Speedup**
ML-KEM / ML-DSA NTT	256	4.2	0.6	7.0x
Ringtail NTT	1024	18.4	2.1	8.8x
Batch 1000 Ringtail NTTs	1024	18,400	2,100	8.8x

Table 5: NTT performance: Intel Xeon 8380 vs NVIDIA A100

8 Chain-Specific Deployment

Chain / Layer	**Primitives**	**Usage**
K-Chain	ML-KEM-768, ML-DSA-65	User transactions, key exchange, contract auth
Q-Chain	Ringtail	Validator threshold ops, collective signing, DKG
Consensus	BLS12-381 + Ringtail	Hybrid dual-path finality [4]
PKI / Governance	SLH-DSA-SHA2-192s	Root certs, constitutional amendments
P2P Networking	ML-KEM-768 + X25519	Hybrid encrypted validator channels

Table 6: Cryptographic deployment across Zoo chains

K-Chain addresses derive as addr = BLAKE3(ML-DSA-65.pk)[0..19]. Archival nodes prune verified ML-DSA signatures after finality, retaining only the signature hash. Q-Chain handles Ringtail DKG at epoch boundaries and threshold signing for cross-chain messages.

9 Formal Verification

All implementations are formally verified in Lean 4:

Theorem 1 (Hybrid Composition Security)**.** *The hybrid scheme* $\Pi_H = (\Pi_C, \Pi_{PQ})$ *with conjunctive verification is EUF-CMA secure if either* Π_C *or* Π_{PQ} *is EUF-CMA secure. Proof: a forger of* Π_H *must forge both components, so* $Adv[\mathcal{A}] \leq Adv[\mathcal{B}_C] + Adv[\mathcal{B}_{PQ}]$*. Full proof in* `Crypto/Hybrid.lean`*.*

File	Property	Lines
`Crypto/MLKEM.lean`	IND-CCA2 security; encaps/decaps correctness; NTT equivalence	1,842
`Crypto/MLDSA.lean`	EUF-CMA security; rejection sampling correctness; no key leakage	2,156
`Crypto/SLHDSA.lean`	EUF-CMA from hash second-preimage resistance; hypertree correctness	1,534
`Crypto/Ringtail.lean`	Threshold unforgeability; DKG correctness; share verification	2,403
`Crypto/Hybrid.lean`	Security holds if either component is secure	687

Table 7: Lean 4 formal verification modules

10 Empirical Evaluation

Signature Scheme	Verify (μs)	Tx Size (B)	TPS (GPU)
ECDSA (secp256k1)	120	245	8,300
ML-DSA-65 (CPU)	250	3,554	4,000
ML-DSA-65 (GPU)	35	3,554	6,200
Hybrid ECDSA+ML-DSA (GPU)	48	3,619	5,800

Table 8: Transaction throughput on Zoo testnet

GPU-accelerated ML-DSA achieves 75% of classical ECDSA throughput with full quantum resistance. Hybrid mode adds 6% overhead. Storage growth (14x per transaction vs ECDSA) is mitigated by signature pruning after finality and separate witness storage.

11 Cross-References and Related Work

11.1 Zoo Network Papers

- **zoo-consensus** [4]: QUASAR hybrid BLS + RINGTAIL finality. This paper provides the cryptographic details referenced there.
- **zoo-key-management** [5]: Key lifecycle, HD wallet derivation for ML-DSA keys, rotation policies.
- **zoo-threshold-signatures** [6]: Extended RINGTAIL DKG, proactive secret sharing, threshold ECDSA for legacy.

11.2 Lux Network Papers

- **lux-pq-crypto-suite** [7]: LUX L0 PQ implementation from which Zoo's suite derives.
- **lux-ringtail-pq** [8]: Original RINGTAIL specification and security proof.

12 Conclusion

Zoo Network deploys a comprehensive post-quantum cryptography suite: ML-KEM for key encapsulation, ML-DSA for transaction signatures, SLH-DSA for high-assurance certification, and RINGTAIL for threshold consensus. Hybrid composition ensures security during the classical-to-quantum transition. GPU NTT acceleration achieves 75% of classical throughput with full quantum resistance. All primitives are formally verified in Lean 4.

References

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